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ROUND TABLE

Organising the unorganised: Role of platform intermediaries in the Indian real estate market

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Abstract A large component of the Indian economy is still in the unorganised (or informal) sector. The emergence of new intermediaries leveraging the Internet has contributed significantly to organising the hitherto unorganised sector. We discuss the case of the Indian real estate industry and elucidate how Internet-based intermediaries have mitigated the problems associated with the unorganised nature of the industry. The three primary problems associated with the unorganised nature of the industry are adverse selection, moral hazard, and weak contract enforcement. Leaders representing four leading real estate platforms discuss how organising the real estate business as a platform business model helps mitigate these risks.
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Academic perspective note

Introduction

A unique feature of the Indian economy has been the prevalence of the unincorporated/non-corporate sector. The unincorporated, according to the legal definition, includes those organisations that are not registered (or incorporated) as independent legal entities, and may include all kinds of firms including agricultural households, self-employed persons, and partnership/proprietorship firms. However, the unorganised sector or the informal sector may be a subset of the non-corporate sector (Vaidyanathan, 2004), and includes non-agricultural production units that "form part of the household sector as household enterprises, or equivalently, unincorporated enterprises owned by households" (NSC, 2012, 9–10). A critical characteristic of the unorganised sector[1] is that the enterprises are not separate legal entities independent of the households/owners; the capital assets used for the business may also be used for other purposes by the owners, and the working capital flows including income and expenditure of the enterprise are inseparable from the cash in hand of the enterprise. In such conditions, it has been a significant challenge

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[1] In this note, we use the terms informal sector and unorganised sector interchangeably.

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to separate out the financial and operational performance of these firms.

The World Bank estimated that the informal economy contributed close to 40% of the Gross National Product (GNP) of low-income countries and 17% for high-income countries (Schneider, 2002). The significance of these unorganised enterprises in the context of the Indian economy is highly accentuated. A 2008 National Commission for Enterprises in the Unorganised Sector (NCEUS) task force sub-committee on evaluating the contribution of the unorganised sector to the GDP estimated that the unorganised sector contribution was over 50% of the Net Domestic Product (NDP) and 91.2% of employment, including over 40% of non-agricultural employment (Mazumdar, 2008; NCEUS, 2008). The National Statistical Commission (NSC) affirmed that the high growth rates of the Indian economy had been accompanied by increasing informalisation (NSC, 2012).

In this note and the accompanying panel discussion, we explore how emerging platform-based business models mitigate the problems of the informal (unorganised) sector and contribute to formalising (organising) the sector. We take the case of the Indian real-estate intermediation market to support our case. The NCEUS (2008) report defined real-estate as a sub-sector that included buying, selling, renting and operating of self-owned or leased real-estate such as apartment buildings and dwellings, non-residential buildings, developing and subdividing real estate into plots, as well as activities with own or leased property on a fee or a contract basis, and estimated the contribution of the unorganised real-estate services (excluding income from ownership) as Rs. 1277 Crores (Rs. 12.27 billion), or about 1% of the GDP (NCEUS, 2008, 24–26).

Issues with the unorganised sector

Farrell (2004) identifies three factors that contribute to the prevalence and growth of the unorganised sector—slack enforcement of regulations, bureaucratic costs of formal operations, and social norms that encourage non-adherence to laws as a tool used by small enterprises to compete with large modern players. Farrel highlights the consequences of operating informally as (i) low productivity levels due to lack of access to capital for investments from the formal financial system; and (ii) the lack of comparable choice for the consumers, with the unorganised sector providing them with cheap, unregulated products and services (Farrell, 2004). Attempts at formalising the informal sector in the past have not had consistent results (Garcia-Bolivar, 2006). Gibson and Kelley (1994) characterised the informal sector as a reserve army for employing those who are not employed/employable by the formal sector. Given the low productivity levels, and the high information asymmetry between the contracting parties, the unorganised sector is likely to be plagued with a variety of contractual issues. Economists studying the unorganised sector highlight three problems with informality—adverse selection, moral hazard, and imperfect contract enforcement (for instance, see Batini, Kim, Levine, & Lotti, 2009; Bell, 1990).

In India, the real estate market had been dominated by unorganised intermediaries, known as brokers. These intermediaries typically operate in local neighbourhoods, use primitive contracting terms (mostly oral contracts), and are one-person operations. A typical customer looking to buy/lease/rent a property would approach these brokers, who would have an inventory of properties in the area that were available. Some of these inventories could be non-exclusively available with multiple brokers as well. Once a broker understands the customer's requirements, he would typically show a shortlist of properties to help make a choice, till such time the customer finds what she was looking for. It is only after that, that the customer (buyer) would be put in touch with the property owner (seller) by the broker, and the broker is paid a commission for the discovery/intermediation. Most often, the broker is thickly involved in the price negotiations, as his commission would be a proportion of the transaction value.

It is in this discovery phase that the problem of adverse selection manifests itself. The seller (and in some cases, the broker as well) may be aware of the problems with the property, which the buyer cannot discover; and hence cannot separate "good" properties from the "bad". The most common issues include unpaid municipal taxes, lacunae in statutory approvals, quality of construction, or even risks of flooding in monsoon rains. The moral hazard problem manifests in the contracting when the broker has an interest in raising the value of the transaction, as his commission is a proportion of the transaction value, rather than acting in the interest of both the buyer and the seller. And finally, when the buyer discovers the real information (that was not present with her when she got into the contract), she has no recourse to make the seller/broker liable, due to poor contract enforcement.

Intermediation platform business models

There are three kinds of intermediation platforms—matching platforms, transaction platforms, and info-mediation platforms. *Matching platforms* are those where customers/users look for one-to-one matches. Typically these are matrimonial websites such as Bharatmatrimony.com. Generally, in order to create an account, users are asked to fill out a questionnaire (Srinivasan & Rao, 2015a). Based on the information given by an individual, the platform provides them with a list of users on the other side with similar interests and sensibilities. The kind of transactions that happen here are one–one and the intent of both sides is for it to be the only transaction they make on this platform, since they hope to find someone and remain married to them for life. Once the transaction is done, the two parties would not engage with the platform any more. A matching platform does not want recurring customers as that would mean that they have not been matched properly the first time. If a customer is unable to find a match and remains on the platform for too long, he/she is considered a bad quality customer by Bharatmatrimony.com, as he/she would be responsible for diluting the quality of the entire platform. It is extremely important for a matching platform to conduct thorough verification on the information provided by the users.

A *transaction platform* facilitates transactions between two sides of the platform, e.g., the buyers and the sellers. It connects the provider of a product/service to the consumer of said product/service. A good example of a transaction platform is Practo.com, where patients can find doctors relevant to all kinds of health disorders and book appointments

with them on the platform (Srinivasan, 2015). A transaction platform needs to have acquired a database of one side of customers such as sellers, or clinics/doctors in the case of Practo, which should be valued by the other side.

An *info-mediation platform* such as Justdial gathers a database of information about one side of customers and provides it to the other side (Srinivasan & Rao, 2015b). Justdial started getting small and medium business(es) (SMBs) onto the platform by listing them free of charge on their site with the promise of augmenting their customer bases, thereby acquiring a database of information up to the street level about these sellers. The kinds of transactions that happen on an info-mediation platform are many. Unlike on a matching platform, both sides of the platform keep coming back to Justdial for more business. A customer who stays on the platform for long is considered a high quality customer in an info-mediation platform.

Matching platform business model

The two sides of a heterosexual matching platform would comprise men and women. There exist strong and positive cross side network effects between the two sides. A larger number of customers on one side implies a broader variety and number of choices for the other side and hence they are more willing to pay for the platform. Negative same side network effects might exist as a larger number of users on their own side would mean less visibility of their profile to the other side.

In a matching platform such as Bharatmatrimony.com, there cannot be a subsidy side of customers. Once a side is subsidised, there would be an incentive for users to provide false information about themselves and moreover there is no indication of whether the user is seriously looking for a potential spouse or has simply created a profile to browse through the site. Moreover, it can become a huge inconvenience for a serious paid user if he/she keeps getting matched with casual unpaid users. Hence, both sides of the platform are money sides for Bharatmatrimony.com and they are charged equally for a subscription. It does give free memberships but it dis-incentivises them as the opposite side would think that the unpaid users are not serious about marriage and/or are price-sensitive. In a matching platform, payment signals seriousness of transaction intent.

Transaction platform business model

A transaction platform acts as an intermediary between the provider and consumer of a product/service. In the case of Practo.com, the two sides of the platform are the clinics and the patients. And there are strong positive cross side network effects between the two sides. As more patients start using the platform to book appointments, a larger number and variety of clinics would want to affiliate with the platform in order to augment their customer base. And as a large variety of doctors working with the clinics are on the platform, more patients would start using Practo.com for booking appointments. Due to the negative same side network effects on the doctors' side, their willingness to pay for premium listings would be high, which Practo can leverage in the future.

Practo.com is subsidised for both patients and clinics. Eventually, the patients will remain as the subsidy side and Practo will start to monetise from the clinics, not through Practo.com but by cross-selling their practice management solution, Practo Ray (Srinivasan, 2015). Unlike in a matching platform or an info-mediation platform, a subscription model does not work here since the number of patients visiting a doctor per month cannot be predicted. Practo cannot charge clinics per appointment because after the first appointment, the doctor could tell the patient to make further appointments with him offline, so that his clinic would not have to pay Practo. So, the ideal model would be to charge clinics for discovery once per customer. However, even this is unlikely to be sustainable, as the number of discoveries may be variable/reduce over time. Therefore, the most sustainable pricing model for a transaction platform is to charge on the volume/value of transactions.

For Practo.com, cross side network effects are strong and positive. Multi-homing costs for doctors/clinics (the money side) are high as they have locked them in with Practo Ray. Practo covered niches like alternative treatment seekers by including Ayurvedic and Homeopathic clinics and doctors on the platform as well. Hence the healthcare app market could become a winner-takes-all market.

Info-mediation platform business model

The two sides of an info-mediation platform are the information providers and the information seekers. In the case of Justdial, the SMBs and the search users make up the two sides. The more the number of search users on the platforms, the more the number of SMBs willing to associate with and pay for the platform. Similarly, the more the number of SMBs on the platform, the more the users willing to search on Justdial to get comprehensive results. Hence, the cross side network effects are strong and positive. However, higher number of SMBs leads to negative same side network effects, which Justdial leverages by charging them for premium listings. It leverages cross side network effects as well by making the SMBs pay for buy leads. Hence, the subsidy side of an info-mediation platform is the search users and the money side is the SMBs.

Justdial, unlike a matching platform, discourages a single seller from paying unusually high because he might be a completely irrelevant seller or a bad seller with deep pockets. Also, it would discourage other sellers from affiliating with the platform. Hence, higher willingness to pay signals different things on these two platforms. If either side of a matching platform is desperate and fragmented, the quality would be poor. In an info-mediation platform, if the SMB's side is desperate and concentrated, then they are good quality customers.

Cross side network effects are strong and positive. All the special features are covered as Justdial is a comprehensive info-mediation platform offering information on a wide range of services and products (Srinivasan & Rao, 2015b). There exist no multi-homing costs for the subsidy side. The SMBs do not require more than one website for an online presence and would not like to pay additional subscription fee for other such platforms. Hence, with first mover advantages and a strong brand image, Justdial has increased the multi-homing costs for the money side. Therefore, the info-mediation platform operates in a winner-takes-all market.

Organising the real estate market—formalising intermediation

A typical real estate intermediation process undergoes four stages—discovery, matching, transaction, and special services.

During the discovery phase, the buyers and sellers (or the lessor and lessee) specify their respective requirements and initiate their willingness to transact. Typically, in the unorganised world, they would reach out to the broker and express their requirements and expectations. This is where, in the absence of formal contracts, significant problems of adverse selection will emerge. Online intermediation platforms like Housing.com ensure that the property listed for sale/lease is supported with the sellers' full contact information, and adequate information about the property including photographs and maps. This begins the mitigation of the information asymmetry in the listing. In order to ensure that there are no perverse incentives for sellers to hide information, a successful platform would not charge money from either the buyers or the sellers. It would provide its plain vanilla, undifferentiated services for free to both sides of the platform, unlike the unorganised broker-driven model, where the broker would charge both sides of users. Such organised platforms would then make money from differentiated services (like assistedproperty.com for buyers; and featured property services for sellers) only. Much like sponsored advertisements on Internet search results, buyers would be aware that these "specially promoted" properties were being paid for to appear above every other result.

During the matching phase, buyers (or lessees) make shortlists (preliminary matches), depending on their requirements and expectations. In the unorganised sector, the broker would play a large part in this match-making process. He would ensure that only those properties in which he has a higher pecuniary interest are shown to the buyers, leading to the manifestation of moral hazard problems. The prioritisation (including the sequence) of properties shown to the buyers would be determined by the broker, as the inventory of properties available in the shortlist is not transparently available to the buyers and sellers. The brokers could use the anchoring and adjustment heuristic (Epley & Gilovich, 2006), and prime buyers towards certain properties (a reference) depending on the buyer's preferences. Intermediation platforms resolve these moral hazards by providing a comprehensive inventory of all those properties that match the criteria defined during the search.

The third and the most critical phase of real estate intermediation is the transaction phase. This is the phase where the buyers and sellers actually transact, and it involves the processes of documentation, registration of contracts, transfer of property rights, and payment of taxes and levies to the local bodies, which entail bureaucratic procedures. Traditionally since brokers would have been in the real estate business for a longer time and deal with these issues regularly, buyers and sellers tend to repose faith in them to ensure smooth transactions. This is the phase where the broker seeks his commission and signs off from the entire contracting process. And, at the end of this process, both the buyer and seller have no recourse to any process in case the contract (or a part of it) is rescinded. This could lead to the manifestation of two significant problems of contracting—risk of holdup, and incompleteness of contract enforcement. Holdup risks could arise when the information asymmetry about the registration and contract documentation process is very high between the broker and the seller/buyer. The broker (the party with high levels of idiosyncratic knowledge) could demonstrate opportunistic behaviour and therefore, lead to value erosion for the buyers/sellers. Given that the broker would typically "vanish" from the process at the end of this stage, the broker has no incentive to ensure that the contracting between the two parties is complete, and therefore could exploit the situation to his advantage. Platform business model intermediaries solve this problem by just being formal organisations, insuring against holdup risks, and ensuring recourse to grievances of either of the parties. Opportunistic behaviour by such formal intermediaries could get a lot of negative publicity and therefore hurt their network effects on either side.

Another significant role of intermediation in the real estate market is the fulfilment of special needs. For instance, an individual might be interested in sharing his apartment with like-minded individuals, and need to discover, contract, and transact with them. In the traditional world, the transactions between house owners and people renting them are one-to-one, and not one-to-many. Here is where we need platforms that can help aggregate, match, and transact the "many" side of the market (in this case, flat-mates). Another example of special services includes management of common areas in a community; this is typically organised by residents' associations and is fraught with problems of volunteering, inefficiencies, and adverse selection. Intermediation platforms like FlatChat help people find compatible flat-mates; and CommonFloor Groups enable these interactions between community members in a transparent and trustworthy manner.

In the panel discussion that follows, we highlight four enterprises in the real estate intermediation market that have significantly contributed to organising the market with their platform business models—Housing.com, IndiaProperty, CommonFloor, and FlatChat.

"Organising" the Indian real estate market: panel discussion[2]

Anchor

R Srinivasan, IIM Bangalore

Panellists

Vikas Malpani, Cofounder and VP, CommonFloor.com
Gaurav Munjal, Founder and CEO, Flatchat[3]
Ganesh Vasudevan, CEO of IndiaProperty.com
Azeem Zainulbhai, CFO, Housing.com[4]

Team Eximius: We are pleased to present the Catalysis Panel Discussion of Eximius 2015, on "Organising" the Indian Real Estate Market. Our panellists for today consist of found-

[2] This panel discussion constituted the Catalysis Panel Discussion at Eximius 2015, the annual entrepreneurship summit of IIM Bangalore. This part of the article carries edited excerpts of the presentations made at the panel discussion. The views expressed by the panellists are personal and academic in nature and not necessarily the views of their organisations. The presentations of the panellists were made in an academic context in an academic institution.

[3] Gaurav Munjal is now Co-Founder and CEO at Unacademy.

[4] Azeem Zainulbhai is now Co-founder and GSD at Intelligent Interfaces.

ers and CFOs of some of the most promising startups in the unorganised real estate sector in the country.

Professor R. Srinivasan from the Corporate Strategy and Policy area of IIMB will be anchoring today's session. The panel members include: Gaurav Munjal, Founder and CEO of Flatchat. Assisting thousands of students and bachelors to find a home, Flatchat helps you find a flat-mate by simply texting, or a feature called Flatchat Assistant. A game changer in the Indian real estate industry, Flatchat has received funding of $2.5 million from CommonFloor and has crossed 45,000+ downloads across both Android and iphone (ioS). On Google Play with the score standing at 4.5, it is the highest rated real estate app with over 1100 ratings. Gaurav is also a founder of Unacademy, a not-for-profit website which hosts educational videos and courses.

Vikas Malpani is the Cofounder and VP of CF Groups CommonFloor.com. CommonFloor Groups is the apartment management offering of CommonFloor.com that provides apartment residents hassle-free community living. During the course of CommonFloor.com's journey, Vikas has been part of the crucial aspects of establishing the company, right from initial sales to campaign management and campaign execution, to product road map and product management. Vikas is also a recipient of the prestigious MIT-Technology Review 35 "Young Innovator of the Year" award.

Azeem Zainulbhai is the CFO of Housing.com. His areas of interest include online real estate classifieds which include fund raising, and building and managing multiple areas within a high growth startup. He has been involved in building and managing a property development company in the Republic of Angola. He is also an entrepreneur having dabbled in voice and data collaboration service for the Indian market as well as online resident reservation. He has also worked in the investment banking sector in Mumbai as well as media and telecom investment banking in New York.

Ganesh Vasudevan is the CEO of IndiaProperty.com, a company that runs and manages an online realty portal. He is a seasoned business dealer with extensive business development experience across roles in sales, strategic alliance, key account management, and direct marketing. He has been particularly successful in seeding and growing green field ventures into large businesses in both B2B and B2C spaces, across Internet banking, financial services and consumer durable industries. Mr. Vasudevan has, during the course of his career, worked with BPL Ltd., Standard Chartered Bank and Tata AIG Life Insurance.

R Srinivasan

The real estate industry has been largely unorganised. When you speak about real estate, the first thing that comes to your mind is the real estate broker and the term "broker" does not always have positive connotations. This industry has also been characterised by transparency issues. However, despite the problems, buying property has been the mark of having settled in life, in the Indian household. It is a critical decision, which takes a lot of time and energy, and involves talking to several influencers, a lot of planning, decision making, and commitment. Several people make other decisions based on their real estate decisions, such as when to get married, whether to postpone their studies to deal with the EMIs and so on. For such an industry to be unorganised was not sustainable, and now the Internet has set up companies which are trying to put some strength into this unorganised business.

We have today a group of four people to describe different aspects of the real estate business. We are going to start with Azeem who will be talking about leveraging the power of data in connecting buyers and sellers. Ganesh is going to talk about how IndiaProperty has been able to not just connect buyers and sellers but also help buyers through assisted property searches, through end-to-end service and find as well as close deals. Vikas is going to talk about what to do within flats—when I move into a flat, how do I commune with the members of the block of flats? Gaurav will talk about how Flatchat helps you find flat-mates and how to get flat-mates and owners together.

Azeem Zainulbhai

I will tell you little bit about Housing.com and about how we leverage data to help buyers through the decision-making process. When we started Housing, unlike other real estate platforms which are focussed mainly on the broker, the middleman in the transaction, we said that the No.1 thing to focus on is the consumer. Let's solve the consumer's problem in buying a house. That took us to a major issue we see in India and globally too, that there are a lot of data missing. There are data transparency issues, data collection issues and data matching issues. There are no right algorithms. We have sets of data here and there which are related but you cannot manipulate them easily. Even when you start assembling these pieces, collecting the data yourself, verifying the data as we do, display of those data is very important. I may have a lot of data but unless the consumer understands how to make a decision based on that data, it is useless.

That is what our product aims to do. We have fantastic data science at Housing that looks at the data we collect, which could be publicly or privately available data, and manipulates or matches that data, which in turn can impact a decision to purchase or to rent. Our product team takes that decision or that influence and packages it in a way that a consumer can actually understand it. For example, on our site, we have something called Life Style that takes into account many different factors and it changes based on the individual's preference. So, when you focus on a consumer, you end up focussing on the data, and when you clarify the data, you end up helping the consumer make a better choice.

From the consumer's perspective, while buying or renting a house, there are many parts to the process. First, the consumer needs to figure out where to go—which is termed as search and discovery. Most platforms initially focus only on search and discovery. Slowly, as more Indian players have come in and revolutionised the industry a bit, we have started to get more data, present the data better, and help the customer with a little more analysis to get him/her closer to the decision-making process. This is a transition from Classifieds 1.0 to Classifieds 2.0. All of us on this panel are trying to reach the Holy Grail of getting into a transaction, that is, helping the customer make a purchase or rent decision.

Our consumer focus and data focus have made us a product and tech company. Our product professionals and our engineers continuously refine and craft the data. Originally, when

there were limited search options, we started collecting our own data. Fake listings are a major problem and we all have different algorithms to extract fake data from our sites. But that is a secondary problem. What it boils down to is that the right data presented in the right way help a consumer make the right decision. There have been cases when a customer goes to a website, sees something nice that he wants to rent, and calls up the number on the site and is told by a broker that it is not available. How do you solve that problem? We are also trying to work on that. However, things have improved over the past two years. If you are a high intent consumer who comes to a property portal in India, right now you will probably have a 25% chance of finding something on that site that you would close on, but two years ago the chances were 5%. So, we have moved 5× in the industry already because we all have better data but there is a long way to go with the world evolving fast.

Ganesh Vasudevan

Three years back, we demerged IndiaProperty as a separate company after we saw the opportunity in the real estate space. At that time the space was pretty much white canvas. Worldwide, and in India too, most models were typical online classifieds 1.0, as Azeem pointed out. There were listings put up by almost anybody and customers looking through these listings hoped to find an answer. At IndiaProperty we tried to figure out which part of the canvas we should start working on. It was a question of prioritisation. When we looked at data across the world, it was quite apparent that last mile data was the most critical factor in the decision-making process and that even in the most developed market such as the United States, a lot of transactions were happening offline as compared to online. Buying or renting an apartment or assessing a builder or a locality is not like buying a cellphone or a pair of jeans where there are easy comparables. You need to get good data on board to help a reasonably informed process of shortlisting and therefore transaction was really important.

When we started, we faced a fully fragmented market and the degree of fragmentation varied between the various transaction types. We had rentals, residential property buyers, commercial rentals, commercial buyers, and institutional brokerage opportunities. These were the broad opportunity spaces which presented themselves at that point. Within these segments, the data tells you that there are some segments which are far more efficient if they remain fragmented—that was our view on rentals. The sheer effort and investment required to collect data on the last mile, to put it up online to enable a rental transaction was too much when the foreseeable revenue was limited to a month or two month's rental as brokerage disintermediation. There was no point in bringing the rentals market onto a platform and organising it.

On the other hand, the property buyer space was far more attractive. The degree of data collection was less involved and there was more standardisation of offerings. For most builders, particularly with new properties, the data are made available upfront, transparent approvals are readily accessible and also easily presentable. The way we reasoned was that since we needed to do a lot of heavy lifting, we might as well do it in areas where we could kick start a revenue stream as opposed to doing it in an area which was not likely to result in any revenue in the foreseeable future. Therefore, we steered clear of rentals and focussed on property buying as a segment. With property buying, you have various sub-opportunities—new property, recent property (in the residential space), commercial property and so on. We diced it further and decided to knock off commercial buying and focus on residential property buys. That is how we built the business.

In 2012, when we rejigged the business model, it was with a focus on new property marketing, specifically helping builders market a little better. But we quickly realised that the true pain point for the consumer in the Indian context is not in discovery, but in transaction. There were several speed bumps that would toss the consumer back several steps in the search process. The time spent on pre-work, on research, and moving closer to the purchase decision was about 6–9 months. Therefore, the challenge was in figuring out how to engage in a meaningful way with prospective buyers and nurture their journey of search and research; it was in helping that buyer figure out between the various alternatives available, and narrow down on the choices in a very scientific way, all the while keeping the lack of quality data at the back of the mind. That would require a hybrid approach, because you have some things given online and some things delivered offline. That is how we evolved and figured out that we need to put in place products and processes which migrate the consumer from online space seamlessly into the offline and back, as and when required, in this search and buy journey. That is how we evolved Assistant Search.

Assistant Search is a paid subscription service where buyers pay a flat fee to help them in the search process. This is a multilayered product; it starts with vanilla search delivered online and subsequently moves into various other elements of products that are to be decided upon when buying a home. This includes providing access to legal opinions, transaction documentation, getting a home loan organised, home insurance, and so on. We quickly evolved into putting all of this into the online ecosystem and the IndiaProperty user can search, evaluate, shortlist, buy only online or offline as the case may be. There have been several technology-based developments such as virtual walk throughs, panoramic walk throughs, live property tours, video reviews of projects, and so on, that have enabled and hastened the path the consumer takes to move to the decision-making point.

From the consumer's point of view, it is very important for us to understand all the pain points. We decided to focus on the biggest one, the actual transaction and the decisions that lead to the transaction. We thought we would focus on that specific area and build up our presence operating in the online and offline space. This included curating content, getting legal opinions upfront, doing a lot of heavy lifting ourselves and putting it up online with a view to monetise this data in a far better way. The business evolved from being a pure online classifieds play to "listings ++" play. We have a set of offerings for the buyer on the demand side which is a subscription-driven product where we handhold the buyer through various stages, and therefore generate highly qualified prospects who are likely to buy. Those prospects are delivered to advertisers and we have been able to generate a far higher premium than our competitors in so far as our ability to monetise is concerned.

On the supply side, we put in place a set of marketing solutions which straddle both online and offline space. For instance, offline fairs are where high value purchase decisions are made; no one with a budget of more than Rs. 75 lakhs (Rs. 750,000) or a crore (Rs. 10,000,000), typically leaves behind their contact details online. They research online and buy offline. The only way to tap into that segment was to provide an offline touch point which would help us to communicate with such customers and monetise to a transaction.

On the other hand, this also helps us to migrate a lot of builders who were not online to start embracing the online medium. The category itself is extremely under penetrated. Therefore, the network efforts are yet to start playing out on both sides of the market place. In that situation, it is important that you help the participants to come out of the market place quickly. The best way to do that is to talk to them in terms that are familiar to them, such as a property fair. They know where this works and that is a great tool for us to get builders on board onto our platform.

So, that is how we put together a complete online plus offline proposition for the supply side of the market; on the demand side we put together an assisted search plus transaction service which is a full stack offering from the consumer side to the buyer side, as well as the seller's side.

Vikas Malpani

I will focus on the journey since the previous speakers have touched on the industry aspect. When we started CommonFloor, we were three men in our early twenties, passionate about solving problems. It began with a problem in the apartment building where one of us lived. The sewage line was disconnected since some arrears had not been paid. What ought to have been a simple matter became a problem that could not be solved. All they had to do was locate the apartment owners, get them together and get the fine paid so that the sewage lines could get fixed. But several of the residents were tenants and the owners, who had to pay the fine, were dispersed all over the world, and could not be contacted. Finally, my friend had to move out of the apartment.

There was a time when important decisions were made in consensus with the family and one's neighbours were part of the extended family and support system. But in a big city, we found that we were living in a society where we were strangers to each other. This was the problem that we first identified. We said let's bring these communities together and build them into the communities of yesteryear. We had no business possibilities, models, or even a business plan, no idea how to make money, but the three of us thought we could move mountains. We thought we would begin by bringing home buyers on a common platform, building these communities, which would be of value to the local businesses, and with these businesses we would advertise and make money.

There were no maps, so we travelled in the areas where we wanted to build communities. I travelled 25,000 km in the first three years to map every single apartment in Bangalore at that point of time! Our first intention was to bring the households together and to get them to communicate with each other. We created a platform, a private network, where they could come only by invite, and talk to each other. We started getting traction as people started finding value in our offerings. There were polls, discussion forums, events, and local task forces that we put together within the apartments.

In this process, we realised that the growth in the number of apartments was making it difficult for the residents or the apartment management committees to run the apartments. The apartments were becoming townships in themselves, with 500 flats and 2000 occupants. We started building systems for them to come together and work on things like management of complaints, tracking payments, accounting solutions—right now we provide the entire building management system. We also realised that we were obsessively focussed on the home buyer. With more apartments coming up, people wanted to rent and buy and resell.

Our first foray into real estate happened when we came up with the most innovative way of looking at the business at that point of time. We did not have listings to share and we were not a real estate player, but we had suppliers. A lot of people who were buyers in these properties wanted to rent out or resell their properties and were exclusively coming to us. Though the word "broker" is not very well received, we realised that there is some value that brokers add. They help closures, specifically with resales. Some of the buyers also wanted to connect for other reasons—either they were not staying on their premises, or did not know what value to ask or how to negotiate, so we started sending these supply leads to them. Our model was not scalable but it was innovative and we had the largest number of communities in the city, even in the country. We realised that there was an opportunity in real estate; we had the best information and data available. And that's when real estate started happening for us.

Our approach was not confined to one model. We said every builder builds a project, which is like a product. Our system was designed as a catalogue of products, and availability within that catalogue, instead of classifieds. In our mind classifieds were meant for items that did not have well defined and standardised attributes, whereas a product has standardised attributes. We started connecting various players and the centre of this ecosystem was the home buyer.

We aimed at building an ecosystem for homebuyers. We had an end-to-end ecosystem of vendors, brokers, developers, owners, people who are looking to rent out, rent in—all of them in one single ecosystem. We adopted the ecosystem approach because we focus on the buyer as an individual and on his needs through his entire journey.

Post buy, we grouped all the home buyer's needs under one group—this included local services, connecting to neighbours, paying maintenance bills, managing security, managing the society, or vendors, for the association. But when we mapped the home buyer's journey, we realised that he faces a lot more pain before he moves into his apartment. At the pre-purchase stage he needs data, and even before that he needs to understand how to buy. We started creating guides for him to understand how to buy.

Most people who want to buy a house get the budget wrong—if your budget is Rs. 50 lakhs (Rs. 5 million), you cannot target a house that costs Rs. 50 lakhs—for that your budget must be at least Rs. 65 lakhs (Rs. 6.5 million). Besides this, there are several other challenges in the ecosystem. The policy from the government perspective is not clear. A company cannot come today and start claiming transparency. You have to start working with the entire ecosystem, all the players,

and start pushing forward. We realised that every player has a place in the ecosystem. There are several good brokers who can help you with right property transactions, and developers doing good constructions and on-time delivery. The question is to identify them. The Holy Grail for us is, after 10 years of living in a society or apartment complex, and working through CommonFloor Groups, if a home owner wants to resell or rent out, or even buy, he comes back to CommonFloor and restarts the journey. This time, he is already pre-equipped and briefed with the information he wants.

Gaurav Munjal

I am sure many of you have heard of the popular online dating app Tinder. While online dating sites have existed for several years, Tinder became extremely popular. On the earlier online dating sites you had to create your profile, look at a bunch of listing profiles and then start selecting the profiles you wanted to see. The rate of response was not high. What Tinder did differently was to find the matches for you, and give you matches, and when both sides approved, you could start chatting. This is an illustration of two different concepts of networking—Do It Yourself (DIY) versus Do It For You (DIFY). Blogger and YouTube also worked on the philosophy of DIFY.

During a chat, it occurred to me that if you can find a date on Tinder, why not a flat-mate? That was the origin of Flatchat.

I believe that in India the rentals space is yet to be fully explored. For example, people used to think that intercity travel is a bigger market than intracity travel. But Uber's revenue in California last year was more than double of what was estimated of the taxi market. That is because once you brought in the concept of shared economy, there were a lot of people who did not use their own cars but used Uber for day to day travel.

A lot of new products are coming in but the rental market is not up for disruption, especially through the shared economy model. The concept of shared economy is now changing. You will not need to buy a car or even own a house. Uber, for instance, changed the uncertainty of travel by connecting you to the right person. The trend in the US is towards renting or even living in shared lodgings such as Airbnbs.

Real estate companies too need to get out of the "listing" approach. While new housing companies have brought in different approaches to rentals, there has been little behavioural change. While brokers might be required for buying a place, for rentals I don't think we would need brokers eventually because of the kind of trends that are emerging. On Flatchat, we saw 400,000 messages being exchanged in the last 30 days between owners and tenants, between tenants who have an extra room and are looking for flat-mates, and between people who want to group up to rent a place. This is an interesting trend because several owners told us that they did not need a broker; they did a profile on Flatchat. Flatchat was similar to Tinder. It showed you five matches and said these are your prospective tenants and these are the requirements that have been taken into consideration, and so on. Then the conversation took off and soon the case was closed. So that is the kind of behavioural change we are talking about. You cannot say that the listing model is a bad model because companies have been using it for the past so many years. What we are saying is that there should be something to disrupt rentals; we are also not saying that we have done it. We are just saying we are headed in the right direction.

With Flatchat, as compared with other companies, it was mobile first. To give you the background, I started working with a software developer called Directi after college, and then I started a company called Flat.to which was a listing site for students and bachelors. However, I was not able to find a flat for myself from my own site. I did not want to build a product that I could not use. That is when I told my investors, let us kill Flat.to. We shut down Flat.to which was seeing hundreds and thousands of users every month. We said we will focus only on Flatchat because that is the next way to go. Page views or search engine optimisation (SEO) does not really matter. Companies like Uber and Tinder do not rely on SEO.

Facebook's business is driven by mobile. That is the trend we are seeing. Forget about mobile first, it's a mobile only country. To buy flats, you need a website but for rental finding, to find flat-mates or flat owners, mobile is the only way to go. With Flatchat, tenants who were looking for a flat-mate created a profile within 10 minutes, could talk to different people and within the same day they were able to find a flat-mate, which takes a lot of time otherwise. And for me, I finally made a product that I could use. It actually helped me find a flat-mate.

Unlike travel or ticket bookings online or e-commerce, real estate is still up for disruption and the rental markets are huge. One of the biggest examples is the success of companies like Airbnb, one of the billion dollar companies across the world, based on the sharing economy.

Discussion

R Srinivasan: We understand from the speakers that there is no need for any intermediary. I want to pose a question to Azeem and Ganesh: Do you think brokers still have value?

Ganesh Vasudevan: They certainly do. It depends on the category and the needs of the customers. The examples that Gaurav gave are more applicable for impulse decisions like choosing an apartment or a flat-mate for a short duration. But while finding a home for a family, and if you want to buy a home, an intermediary certainly helps. This may be less relevant for rentals. As Vikas pointed out, there are a lot of brokers who do a truly good job of presenting actionable neutral information. There is significant value addition that happens. However, as pointed out earlier, there are segments which are more efficiently served if left fragmented and where the shared economy concept is unlikely to work. You cannot paint all segments with the same brush. There are different categories; rentals, sharing a room, and buying a home are three different need states altogether. The information needs are different for the consumer at that point.

R Srinivasan: Azeem, would you like to talk about how rentals work? About disintermediation?

Azeem Zainulbhai: It takes a lot of courage to say right away that we are going to disintermediate. At some points, you can effect that change and at others, you cannot.

From our perspective we know that there are certain brokers that are high quality and who actually do help, even in rental cases. At the end of the day we care about the

customers, so if we are going to solve the problem, our brokers are going to play a part in that. Good brokers are great, when they can solve the customer's problem. But if they are bad brokers what do you do? Do you get rid of them or make them better? It is not a binary choice. Some of them will not come to your platform because they do not have the requisite data. But a part of what you create can also be tools to help them out. You have to take a step in the right direction to change the system.

R Srinivasan: We understand that the CommonFloor ecosystem has an aggregate of several small brokers. An offline broker could well have a good inventory of a small part of the geography of a city. CommonFloor, by aggregating the brokers, is increasing the economy itself, making the inventory large.

Vikas Malpani: We believe that real estate is a hyper local plane. In the process of data aggregation and information gathering, it takes a lot of time to reach the intelligence of getting the hyper local data. The brokers that exist in different markets are the ones who have the intelligence with them. Our system works with them both offline and online. When I say system, it is about the process and the product. We work with them to get to that part where intelligence can come into the system and get showcased. At the end of the day, a lot of things about making a large decision are about human feelings. A buy decision is a big decision. It is a lot about how you feel about that location and the broker plays an important part. By aggregating all the brokers and trying to get through the process, we are also making sure that each of these regions has the right representation in the right people. The demand–supply mismatch is huge and the entire ecosystem is pretty dynamic. In the last 10 years, in Bangalore for instance, a lot of people have moved to Whitefield and Sarjapur. The demand–supply earlier was towards MG Road, Brigade Road, Rajajinagar, and so on. The ecosystem trains itself while you are moving it through the process and the product. The overall aggregation happens and through the algorithm that we have built, our objective is to slowly bring to the surface better people and better properties.

We are also working with brokers to help them understand the value of our processes. They are professionals and we have to start recognising them as professionals. Through our system we are trying to identify and train these professionals. Finally you will have high quality professionals instead of "brokers". I will say that they are consultants who will help you with your needs, if you choose.

R Srinivasan: Gaurav, your argument is based on the idea that rental is a commodity and I believe there is some truth in it. At the end of the day, it is a 10×10 room and I am going to spend 8 to 10 hours in it and not more; but there is also a feeling attached to the whole thing. I need to take time to make the full decision. Do you think the time taken to take the full decision is coming down? Is there a commoditisation of rentals?

Gaurav Munjal: As a background to one of the new features we are coming up with on Flatchat, we noticed that more and more people are messaging on their mobiles rather than calling up. But the same questions are being repeated again and again such as questions on the distance from certain landmarks, the rental policy on bachelors and so on. So, one of the new features our team is coming up with is that when you create a profile, you provide a one line message about your place. When a customer sees your profile and clicks on play, he/she can hear what you have to say. It is like sending voice messages on WhatsApp. Things like this one minute pitch were not possible on the Web—they are much easier on the mobile where you just click on the microphone and you record a one minute message. People are already using this on WhatsApp and Messenger. If a feature like this is available on Flatmate will the time taken to decide come down or go up? I think the answer is self-evident.

Audience Question: Our population is around 120–130 crores (1200–1300 million) and out of that, smart phone users number 20–30 crores (200–300 million). And you are fragmenting your target group? Are you targeting a very small part of the market?

Gaurav Munjal: The number of smart phone users has the highest level of growth that you can imagine. And it is not just us, every segment around us is going in that direction. Travel companies, companies like Flipkart and so on, are promoting their apps. There are certain kinds of innovation that are only possible if you focus on mobile. I want someone to buy a phone to use Flatchat.

Audience Question: What about the reusability of your app?

Gaurav Munjal: That is the problem we are trying to solve. The uninstall ratio is huge at 75–80%. I don't think I have an answer to that.

Audience Question: How do companies such as the ones on this panel increase their visibility?

Azeem Zainulbhai: It is a question of marketing. If you want people to know about you, in this context, the best way possible is through word-of mouth (WOM) spread, whereby everybody knows about you and uses your product.

But in an industry like ours, people make a purchase decision infrequently; rentals may have a higher frequency. So my concern is, if somebody is using Housing today how do I make sure that we are on top of that person's mind three and a half years from now. A part of the answer to that is offline marketing, such as big billboard campaigns. Another part of it is to just have an awesome product so that people talk about you. Another way is to find out how people search for their house on search and discovery platforms like Google or Facebook, and you need to show up there.

Vikas Malpani: To add to that, we must understand that we are in a state of metamorphosis. There is a lot of on-ground effort going on, but with the efforts of all the players put together, we represent not even 5% of the transactions happening online right now. The market is huge. In India, it is an $80 billion residential real estate market and I am not even including rentals. The market is hyper local. In the last few years we have been organising the ecosystem piece by piece and building it without a policy framework or data. As you start moving forward you will start seeing better products in the market and more people using this system as their primary go-to. The next generation is going to fuel the real estate online ecosystem.

Ganesh Vasudevan: This is a classic chicken and egg situation. You build a product and a business around it and then build scale. There are two paths you can take. The top one is what Flatchat is doing in terms of building a great product hoping that usage, WOM, will drive it and bring it to a point of scale which excites people to invest (which is what CommonFloor has done) and use those dollars to build the

brand even more—this becomes a self-fulfilling prophecy. Then you take out a full front page ad in a major national newspaper and people will start talking about you. This will take a few crore rupees. To reach that stage, you will need to do the hard work much earlier. Each of our products is at different stages of evolution. From that perspective there is also a target group (TG) match. What we try to do is be really sharp and focussed in marketing, use property value and speak only for your specific TG. Invariably you see that visibility, recall, top of mind is a direct function of your share of voice, and share of voice is a direct function of the marketing dollars that you pump in, which is a function of how well your product and your business are doing and how well you have been able to monetise. So all these need to fall in place.

Gaurav Munjal: One of the things we have realised in the past two years is that tech is easy. To build a product, to get a developer is easy; but marketing is a difficult game. In my opinion, if you have a mobile product you do not need billboards and TV campaigns. One has never seen Tinder hoardings. Uber has not put up hoardings or done a TV campaign in the US and yet they are disrupting the whole market. What they do is they give you a free 300 rupee ride. If your product is great and if you can get someone to use it for the first time, what you need to do is make it addictive. The first time I used Uber, at the click of a button the driver came to my doorstep and that was a free ride, and I realised that my life was going to change after this. You make an awesome product and give one part of it to the user to use, because if it is an awesome product it will grow, or else it won't.

Audience Question: My question is to Azeem about the real estate product Housing. You have a lifestyle rating to your application which provides more granular data to your customers. How do you build trust in your customers to believe that lifestyle rating? Every customer will have a different point of view from your lifestyle rating. How do you manage to keep up the ratings? How many people do you deploy to see whether the house in question is up to the ratings?

Azeem Zainulbhai: There is a classic way that we build our promise. Our data scientists with say 180 pieces of data set up an algorithm that will turn these 180 pieces of data into one number or one ranking. The first time we did it, it did not work very well. We got negative feedback from people. Through rounds of iteration, the product got better. At first, it was static algorithm where everybody got the same rating but now we are smart enough to know that certain users and certain behaviour will move towards different pieces of the data being more important, so the actual weightage of the data changes dynamically. Are we there yet? No, I don't think so. I feel there is more that has to be done to understand the user better. We continue to learn over time. Our data scientists are constantly working on these algorithms and fixing them.

Audience Question: Why don't you let your customers rate the product so that you have your own crowd sourcing data?

Azeem Zainulbhai: We have the Indian Real Estate (IRE) forum that does community rankings. We are using that data to get better rankings and we are rolling out a product feature soon that will allow customers to give their rankings.

R Srinivasan: In a very infrequent transaction, a Zomato kind of a review rating system may not be very useful and credible…

Ganesh Vasudevan: That is a very valid point. Given this category, there can be no true currency that can be built on any crowd sourced data. We have ratings on IndiaProperty crowd sourced from people staying in that locality. Yet the utilisation of that is always followed by a question mark. We have to evolve a common currency which can be ratified by government agencies where people can understand and relate to the rating.

Audience Question: Your revenue model is something that most people are not aware of. How are you planning to get revenue from your products?

Azeem Zainulbhai: Interestingly, when we were fund raising, we were already thinking about monetisation. Our advisers believed that you should hit 80% market share before you go out and monetise. That is one point of view. But, we are not in the market place business. What we believe and what we did was, in the first year, our focus was very deep on the profit; the second year was on supply; the third year was deep on demand—which was why you saw our big marketing campaign. What we are focussing on now is monetisation. How do you make money in these businesses? You allow people to advertise on your site, you allow people to transact on your site. Then, in between, you can do a leads based model, as in I give you a lead and you pay me, or you do a subscription-based model. Currently, on our platform we do a little bit of advertising only for developers; it is not for rent. We do transactions; people pay money to buy a house on our site.

Gaurav Munjal: But why do we need *to make* money? I think Wikipedia was not making money for a long time but because they had a product that people were using, they asked people to pay. Our investors too were asking us how we would make money and we had to show them. So we conducted an experiment for three weeks. We had an assistant with whom the customer could also chat, and which would reply in five minutes. We told our users that if you find our app useful, leave us a tip.

And we had hundreds of users who paid us. We did not even say it was compulsory. Later, if we want to render into the premium version, we might do something. I remember for a week around 320 people paid. They left a tip of Rs. 299 each. We were saving them a lot of money in broker's fees. So if you are deep into the market, it won't be difficult to make money. But I believe that money should come from the consumer. You must make a product that consumers are willing to pay for and the money must not come through brokers or promotion or advertising. I believe that if you make an awesome product, consumers will pay.

R Srinivasan: Thank you all for being here and participating in this discussion.

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